Fiscal Year:	FY 2009	**Task Last Updated:**	FY 07/10/2009
PI Name:	Rose, Raphael Ph.D.		
Project Title:	A Multi-Media, Computer-Based, Self-Directed, Autonomous, Stress and Anxiety-Management Countermeasure		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Neurobehavioral and Psychosocial Factors Team		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rose@psych.ucla.edu	**Fax:**	FY 310-825-9048
PI Organization Type:	UNIVERSITY	**Phone:**	310-825-9048
Organization Name:	University of California, Los Angeles		
PI Address 1:	Department of Psychology		
PI Address 2:	Box 951563, 1285 Franz Hall		
PI Web Page:			
City:	Los Angeles	**State:**	CA
Zip Code:	90095-1563	**Congressional District:**	33
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2007 Crew Health NNJ07ZSA002N
Start Date:	07/01/2008	**End Date:**	06/30/2012
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Buckey, Jay (Dartmouth College) Cartreine (nee Carter), James (Beth Israel Deaconess Medical Center) Craske, Michelle (University of California, Los Angeles) Beven, Gary (NASA JSC) Glover, Dorie (Dorothy) (University of California, Los Angeles)		
Grant/Contract No.:	NCC 9-58-NBPF01605		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Stress and anxiety-related problems before, during and after space missions can seriously compromise crew efficiency and safety. The Institute of Medicine report, of Earth-analog environments (e.g., Antarctica) suggests a prevalence rate of serious behavioral problems of up to 13% per person per year, highlighting the potential behavioral problems that could occur on a 3-year mission. Further, the potential severity of this problem is highlighted by Bioastronautics Roadmap Risk #25 - human performance failure due to mood alterations such as depression, anxiety, or other psychiatric and cognitive problems - a Priority 1 risk for ISS and Mars. Our project has two specific aims. The first aim is to develop a self-directed, autonomous, interactive multimedia program to train astronauts how to recognize, assess, detect, prevent, and manage stress and anxiety on extended spaceflights. The countermeasure would be principally used in pre-flight training, with booster sessions or interventions available on orbit as needed. In this model of implementation, the skills learned prior to a mission would help to "inoculate" fliers to severe stress reactions thereby enhancing flight-task performance and crew health, safety and efficiency. The countermeasure will use empirically supported stress and anxiety management strategies delivered by interactive media in the NSBRI-supported Virtual Space Station suite of psychosocial support programs. The second specific aim is to evaluate the efficacy of the program with sample of Navy Pilots (who are similar to astronauts) in a randomized controlled trial. The final product will be a computer-based stress management program that could be used both by flight crews and ground-personnel when needed both during training and during missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	At the end of this project, we will not only have developed a countermeasure with potential value to the space program, but also to the American public. Stress-related health and mental problems are among the most common and costly in the country. Although large-scale testing in the public is needed after this project's completion to further identify in which sub-populations and user settings (home, clinic, etc.) it is most useful. The stress management program has the potential for widespread dissemination via broadband Internet to provide stress-management treatment to individuals worldwide.
Task Progress:	In the first year of this project our research team has developed the content of our computerized program called Stress Management and Resiliency Training (SMART). The SMART manual details the scope and focus content of the stress management computer countermeasure. The SMART manual was created by team experts (Drs. Rose, Craske, Cartriene, and Buckey). The manual was reviewed by team consultants and collaborators such as Dr. Donald Meichenbaum, creator of Stress Inoculation Training, long-duration astronaut, Leroy Chiao, and Drs. Gary Beven, Al Holland, Walter Sipes, Lauren Leveton, and Camille Shea for astronaut accuracy and acceptability. Drs. Rose, Cartreine and Buckey presented the aims of the research project to members of the US Naval Center for Combat & Operational Stress Control (NCCOSC) team in San Diego, CA. We formalized a cooperation agreement with researchers at NCCOSC to conduct our efficacy trial (YR 3 and 4) with a sample of US Naval pilots. In June our team will be presenting information about our SMART manual to a group of 10 Naval personnel to get qualitative feedback on our approach. Our team is currently developing the design document that details the look and structure of the stress management countermeasure. Multimedia production has begun on the stress management assessment and detection module and scripts for the introductory greeting of the program are under development. Design aspects for incorporating the stress management program into the existing VSS are being addressed with team members and multi-media professionals. Ongoing regular communication via teleconferences and in person meetings with key investigators and collaborators occur to help plan and monitor progress towards team goals.
Bibliography Type:	Description: (Last Updated: 02/11/2021)
Awards	Rose RD. "Distinguished Faculty Teaching Award, Non-Ladder Faculty, UCLA Dept of Psychology, May 2009." May-2009